Excitation-Preserving Distributed Safety Filter for Multi-Agent Adaptive Control on Complex Dubins Vehicles

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***Abstract*—For a network of N complex-Dubins agents with unknown loss-of-effectiveness (LOE) on the turn-rate channel λ_i, we identify the convergence rate of the parameter estimate $\hat{\theta}_i \to 1/\lambda_i$ under a distributed higher-order CBF (HOCBF) safety filter as the constrained scalar Fisher information of $1/\lambda_i$ on the binary freedom cone $F_i \in \{\{0\}, \mathbb{R}\}$—equivalently, the inverse of the corresponding scalar Cramér–Rao bound. The lift from the v16 single-integrator construction to the v17 complex-Dubins setting introduces (i) relative-degree-2 safety constraints requiring the second-order Aubin–Cellina tangent cone, (ii) a relative-degree-drop locus D_{ij} where Filippov regularisation gives sliding-mode evolution, and (iii) a quantitative recoverability margin $\eta_a > 0$ guaranteeing forward invariance under saturation. The construction composes eleven pre-1985 classical objects: Nagumo's viability theorem (1942) lifted via Aubin–Cellina (1984) §5.1 second-order tangent cone, Filippov (1960) regularisation, the Moreau (1965/1971/1977) proximal-operator/sweeping-process triplet, Brezis (1973) maximal monotone evolution, Noether (1918) gauge symmetry, Hilbert–Courant (1924) min–max, Krasovskii (1959) ultimate boundedness, Krasnosel'skii–Pokrovskii (1989) play operator, Birkhoff (1931) ergodic theorem, Klein (1872) Erlangen + Rao (1945) constrained Fisher information, and Tikhonov (1952) singular perturbation. The OSQP scalar QP at $h_{\text{outer}} = 5\,\text{ms}$ is real-time feasible up to $N \approx 4$ on a desktop CPU; communication-delay robustness improves $20\times$ over v16 *at* $N = 4$ ($5\,\text{ms} \to 100\,\text{ms}$), due to the analytical R3 latency residual in the CBF tightening $\delta_{ij}(t)$. At $N = 8$ the active-set indicator's BV total variation rises proportionally and $\delta_{ij}(t)$ partially absorbs broadcast delay: $\tau \le 5\,\text{ms}$ preserves $h_{ij} \ge 0$; $\tau \in [20, 100]\,\text{ms}$ produces $|h_{ij}^{\min}| \le 0.07\,\text{m}^2$ absorbed by the Tikhonov soft slack — quantified separately in §VII.**

I. Problem statement

A team of N complex-Dubins agents must reach a target formation $\{t_i\}_{i=1}^N \subset \mathbb{C}$ while (i) maintaining pairwise safety $|r_i - r_j| \ge r_{\text{safe}}$, (ii) identifying each agent's unknown LOE gain $\lambda_i \in [\lambda_{\min}, 1]$ on the turn-rate channel, and (iii) using only neighbour-broadcast information $\{r_j, v_{a,j}, \hat{\theta}_j, u_{2,j}^{\text{safe}}(t^-)\}$ from a connected communication graph $\mathcal{G}$. We extend the single-agent adaptive-Dubins formulation of Maldonado-Naranjo–Annaswamy (2025) [1] to the multi-agent setting with HOCBF safety constraints (Xiao–Belta 2021) [2] and prove the convergence rate of $\hat{\theta}_i$ as the constrained Cramér–Rao bound on the binary freedom cone (Rao 1945 [3], Whitney 1965 stratification [4], Amari 1985 information geometry [5]).

II. Construction

A. Plant, reference, adaptive law, Kalman–Bucy auxiliary

For agent $i \in \{1, \dots, N\}$ with state $(r_i, v_{a,i}) \in \mathbb{C}^2$, position $r_i = x_i + iy_i$, velocity-along-heading $v_{a,i} = V_0 e^{i\psi_i}$ at constant speed V_0:

$$\dot{r}_i = v_{a,i}, \qquad \dot{v}_{a,i} = i\,\lambda_i\,u_{2,i}\,v_{a,i},$$

with scalar real turn-rate $u_{2,i} \in \mathbb{R}$. By Carathéodory (1909) accessibility [6] and Chow (1939) bracket theorem [7], the system is reachable on $SE(2)$ via $u_{2,i}$ alone—the Lie bracket of the forward-motion vector field $X_1 = v_{a,i}\partial_{r_i}$ and the turning vector field $X_2 = iv_{a,i}u_{2,i}\partial_{v_{a,i}}$ generates the missing lateral direction.

The reference, adaptive law, and KB filter take the complex-regressor form (paper §2):

$$\begin{aligned}
u_{2,i}^{\text{ref}} &= u_{2,\text{ref},i} - \Re\Big(\frac{K_T e_i + K_F \sum_j [(e_i - e_j) - \cdots]}{i\,v_{a,i}}\Big),\\
u_{2,i}^{\text{AC}} &= \hat{\theta}_i\,u_{2,i}^{\text{ref}}, \quad \tilde{v}_i := v_{a,i} - v_{\text{ref},i},\\
\dot{\hat{\theta}}_i &= \text{Proj}_{[\theta_{\min},\theta_{\max}]}\Big[-\tfrac{\gamma}{m_i^2}\,\Re\big(\overline{iv_{a,i}u_{2,i}^{\text{ref}}}\,\tilde{v}_i\big)\Big],
\end{aligned}$$

with $m_i^2 := 1 + V_0^2(u_{2,i}^{\text{ref}})^2$, the Pomet–Praly normalisation (Pomet–Praly 1992) [8]. The KB filter $\dot{P}_i = -P_i^2|v_{a,i}u_{2,i}^{\text{ref}}|^2/m_i^2$ runs in parallel; under closed-loop PE, $P_i \to 0$ exponentially (Anderson 1985 [9]; modern Robbins–Monro / Borkar 2008 framework [10], [11]).

B. HOCBF + Filippov regularisation at the locus

The pairwise CBF $h_{ij}(r) := |r_i - r_j|^2 - r_{\text{safe}}^2$ has **relative degree 2** w.r.t. u_2 (Aubin–Cellina 1984 §5.1 second-order tangent cone [12]; Xiao–Belta 2021 HOCBF [2]). Define $\psi_{1,ij} := \dot{h}_{ij} + \alpha_1 h_{ij}$; the HOCBF condition $\dot{\psi}_{1,ij} + \alpha_2\psi_{1,ij} \ge 0$ multiplied through by $\hat{\theta}_i$ yields the gauge-fixed constraint

$$\begin{aligned}
c_{ij}(u_{2,i}; r, v_a, \hat{\theta}) := {} & 2\,u_{2,i}\,\Im\big((r_i - r_j)\overline{v_{a,i}}\big)\\
& - 2\tfrac{\hat{\theta}_i}{\hat{\theta}_j}u_{2,j}^{\text{safe}}(t^-)\,\Im\big((r_i - r_j)\overline{v_{a,j}}\big)\\
& + \hat{\theta}_i\,b_{ij}^{(0)} \;\ge\; \delta_{ij}(t),
\end{aligned}$$

with $b_{ij}^{(0)} = 2|v_{a,i} - v_{a,j}|^2 + 2(\alpha_1 + \alpha_2)\Re\big((r_i - r_j)\overline{v_{a,i} - v_{a,j}}\big) + \alpha_1\alpha_2 h_{ij}$ the corrected $\hat{\theta}$-independent residual.

Relative-degree-drop locus. At $\arg(r_i - r_j) - \arg v_{a,i} \in \{0, \pi\}$, the HOCBF coefficient $a_{ii} = 2\Im\big((r_i - r_j)\overline{v_{a,i}}\big)$

vanishes; the differential inclusion becomes non-Lipschitz. We adopt **Filippov (1960)** regularisation [13]: the convex-hull set-valued vector field gives an absolutely-continuous solution with sliding-mode behaviour along D_{ij}. The slack-penalty QP is the smoothed Tikhonov approximation (Tikhonov 1963 / Engl–Hanke–Neubauer 1996 [14], [15]) with $\mathcal{O}(M^{-1/2})$ rate.

Recoverability margin (paper §3.1.1). Axiom (A3") requires $|a_{ii}| \geq \eta_a > 0$ on the closed-loop invariant set, with η_a chosen so $\dot{\psi}_{\max} \cdot \eta_a \geq |\text{res}_{ij}|_{\text{worst}}$. For the §8.3 parameters, $\eta_a^{\text{practical}} \approx 1.6$.

$\delta_{ij}(t)$ **residual aggregation (paper §3.1.2).** Three sources—(R1) gauge-fix LHS-coefficient residual, (R2) cross-term substitution residual, (R3) broadcast-latency residual—aggregate into

$$\delta_{ij}(t) = \dot{\psi}_{\max}[|a_{ii}|\varepsilon_i^{(1)} + |a_{ij}|\varepsilon_{ij}^{(2)}] + \tfrac{\hat{\theta}_i}{\hat{\theta}_j}|a_{ij}|\varepsilon_j^{(3)}(\tau_d) + \zeta_0,$$

where $\varepsilon_j^{(3)}(\tau_d) = L_{\text{QP}} \cdot \tau_d \approx 0.0094$ at $\tau_d = 5\,\text{ms}$—the analytical absorbing of broadcast latency that yields the $20\times$ communication-delay-robustness improvement over v16.

C. QP-resolvent (Moreau prox + KKT closed form)

The closed-loop control is the Moreau proximal operator (Moreau 1965/1971/1977 [16]–[18]) on the time-varying interval $K_i(t) \subset \mathbb{R}$, satisfying the prox-regularity hypothesis (H-PR) (Edmond–Thibault 2006 [19]). The safety filter is a Maxwell-governor (Maxwell 1868 [20]) on the AC command:

$$u_{2,i}^{\text{safe}} = \arg\min_{u_2, s \geq 0} (u_2 - (u_{2,i}^{\text{AC}} + \tilde{e}_i^{\text{pe}}))^2 + M \sum_{j \in \mathcal{N}_i^{\text{on}}} s_{ij}^2$$

$$\text{s.t. } c_{ij}(u_2; \cdots) + s_{ij} \geq \delta_{ij}(t), \;\; s_{ij} \geq 0, \;\; |u_2| \leq \dot{\psi}_{\max}. \quad (1)$$

For $|\mathcal{N}_i^{\text{on}}| = 1$, the closed-form Karush–Kuhn–Tucker (KKT, Karush 1939 [21] / Kuhn–Tucker 1951 [22]) solution is $u_{2,i}^{\text{safe}} = (u_{2,i}^{\text{AC}} + \tilde{e}_i^{\text{pe}}) + \mu_{ij}^* a_{ii}$ with $\mu_{ij}^* = \max(0, (\delta_{ij} - c_{ij})/a_{ii}^2)$, satisfying complementary slackness $\mu_{ij}^*(c_{ij} - \delta_{ij}) = 0$. For $|\mathcal{N}_i^{\text{on}}| \geq 2$ (multi-pair active set), no closed-form Lagrangian exists; OSQP solves the QP numerically. The exact-penalty threshold is $M^* > \|\lambda^*\|_\infty$ (Bertsekas 1999 §5.4 [23]); empirically M scales as $\sqrt{N(N-1)/2}$ with the active-set cardinality.

Empirical OSQP timing (Boyd [24], Passes 31/37). At $N = 4$ (6 pairs): 0.31 ms warm / 5.3 ms cold. At $N = 8$ (28 pairs, $\text{K}_8 antipodal - ringrosette, worst - caseall7neighboursactiveforoneagent$) :$\text{p}_{50} = 0.343\,\text{ms}$, $p_{99} = 0.486\,\text{ms}$ warm-started; 2.74 ms cold-start. Serial 8-agent p_{99}: 3.89 ms / $h_{\text{outer}} = 10\,\text{ms} = 38.9\%$ utilisation (well within budget). The §VI N=8 demo bumps h_{outer} $5 \to 10\,\text{ms}$ (Tikhonov margin $\varepsilon = K_T \Lambda_{\min} h_{\text{outer}} = 0.024$, separation factor $42\times$); §VI's $N = 2$ proof-bearing example keeps the baseline 5 ms.

III. Symmetry, Lyapunov, identifiability

A. Gauge $\mathbb{R}_{>0} \times U(1)$ + shape-space reduction

The Lyapunov function $V = \sum_i V_i + V_{\text{form}} + \kappa \sum_{i<j} B(h_{ij})$ with $V_i = (|e_i|^2 + |\tilde{v}_i|^2)/(2m_i^2) + (\lambda_i/2\gamma)\tilde{\theta}_i^2$ (Noether-required $\kappa_v = 1$, Lie 1880s [25]) is invariant under the Klein-Erlangen gauge $\mathbb{R}_{>0}$ on $\hat{\theta}$ and the Weyl gauge $U(1)$ on heading rotation (Klein 1872 [26], Weyl 1918 [27], Noether 1918 [28]). The cross-swap (§VI) is a **rotational relative equilibrium** (Leonard et al. 2007 [29]) of the reduced shape-space dynamics on $SE(2)^N/U(1)$.

B. Lemma 2 (Krasovskii UUB, v17 form)

$\dot{V} \leq -\eta(\sum_i(|e_i|^2 + |\tilde{v}_i|^2)/m_i^4 + \|\xi\|^2) + \mathcal{O}(A_e^2 + \sup_i P_i^2 + M^{-1/2})$, with $\eta \geq \min(K_T/2, \lambda_{\min}/2 - \bar{\mu}^2 L_{\text{QP}}^{*2}/(2K_T))$ via Pomet–Praly cancellation (Noether-conserved current of G_κ), Hilbert–Courant min-max (1924) [30] on the **Kirchhoff Laplacian** $L_{\mathcal{G}}$ (Kirchhoff 1847 [31], Fiedler 1973 algebraic connectivity [32]), and Tikhonov (1952) singular perturbation [33] for the heading-PD inner-loop cascade (Maldonado-Naranjo–Annaswamy 2025 §III.B [1]). At §8.3 parameters: $K_T \lambda_{\min} = 2.4$ vs threshold $\bar{\mu}^2 L_{\text{QP}}^{*2} \approx 0.32$—margin $7.5\times$.

C. Birkhoff identifiability + scalar Fisher info

The binary freedom cone $F_i \in \{\{0\}, \mathbb{R}\}$ (cruising vs. active-pair) Whitney-stratifies the closed-loop measure μ (Whitney 1965 [4], Krasnosel'skii–Pokrovskii 1989 hysteresis [34], Davis 1984 PDMP [35]). The constrained scalar Fisher information of $1/\lambda_i$ is

$$\boxed{\bar{\rho}_i := \mathbb{E}_\mu\Big[\tfrac{V_0^2(u_{2,i}^{\text{ref}})^2}{1+V_0^2(u_{2,i}^{\text{ref}})^2} \cdot \mathbb{1}\{\mathcal{N}_i^{\text{on}} = \emptyset\}\Big] \in [0,1).} \quad (2)$$

This is the Amari (1985) mixture-coordinate framing on the Whitney-stratified parameter manifold (Fisher 1925 [36] + Rao 1945 §6 [3] + Stoica–Ng 1998 engineering CRB [37]). The convergence rate is $\rho_i = \gamma\bar{\rho}_i$.

IV. Theorem

[Excitation-preserving distributed safety filter, complex Dubins, v17.1] Under axioms (A1) projected $\hat{\theta}_i \in [\theta_{\min}, \theta_{\max}]$, (A2)+(A2') closed-loop PE, (A2") free-time dwell, (A3") HOCBF initial conditions $h_{ij}(0) \geq \delta_{ij}(0) + \zeta$, $\psi_{1,ij}(0) \geq 0$, $|a_{ii}| \geq \eta_a > 0$, (A4) continuous broadcast of $\{r_j, v_{a,j}, \hat{\theta}_j, u_{2,j}^{\text{safe}}(t^-)\}$, and (A5) active-set non-saturation ($|\mathcal{N}_i^{\text{on}}| \leq 1$), the closed-loop trajectories of $\dot{r}_i = v_{a,i}$, $\dot{v}_{a,i} = i\lambda_i u_{2,i}^{\text{safe}} v_{a,i}$, $|v_{a,i}| = V_0$ generated by Moreau's (1971) sweeping process off the locus D_{ij} and Filippov's (1960) regularisation on D_{ij} satisfy:

(1) $h_{ij}(r(t)) \geq 0$ for all $t \geq 0$, by Nagumo's viability theorem (1942) [38] lifted via Aubin–Cellina (1984) §5.1 second-order tangent cone [12], with sliding-mode behaviour on D_{ij};
(2) ultimate boundedness $V(t) \leq V(0)e^{-\eta t} + \mathcal{O}(A_e^2/\eta) + \mathcal{O}(\sup_i P_i^2/\eta) + \mathcal{O}(M^{-1/2}/\eta)$ [39];
(3) scalar parameter convergence $\hat{\theta}_i \to 1/\lambda_i$ exponentially with rate $\rho_i = \gamma\bar{\rho}_i$, where $\bar{\rho}_i$ is the constrained scalar Fisher information (2).

V. Proof outline (six lemmas, all classical)

1) **Lemma 5.1 (forward invariance).** Aubin–Cellina + Filippov + Krasovskii reverse-Lyapunov on $\psi_{1,ij}$. On-locus dwell-time (Pass 33 re-derivation): $\tau_d \geq \varepsilon_{\text{hyst}}/(L_{\text{eff}}|c_{ij}|'_{\max}) \gtrsim 10^{-3}$ ($\sim 1\,\text{ms}$ physical at §8.3 parameters; D_{ij} is μ-measure-zero).
2) **Lemma 5.2 (estimation tolerance).** R1+R2+R3 residual aggregation; $\delta_{ij}(t) \to \zeta_0$ as $P_i, P_j \to 0$ (Anderson 1985 [9]).
3) **Lemma 5.3 (Krasovskii UUB).** As §III.B above.
4) **Lemma 5.4 (resolvent well-posedness).** Moreau prox + Brezis 1973 [40] + Filippov on D_{ij} + Hager 1979 [41] Lipschitz with $L^*_{\text{QP}} = (1+\kappa_\lambda)/\eta_a$.
5) **Lemma 5.5 (Birkhoff identifiability).** Birkhoff 1931 [42] ergodic theorem on the K-P / PDMP invariant measure μ; $\bar{\rho}_i > 0$ under (A2')∩(A2").
6) **Lemma 5.6 (uniform multi-pair dwell-time).** K-P 1989 BV continuity + Hager 1979; $\tau_d > 0$ uniformly.

VI. Worked examples

A. $N = 2$ *head-on (Filippov sliding mode demo)*

For $r_1(0) = -L$, $r_2(0) = +L$, $v_{a,i}(0) = \pm V_0$ along the real axis: $a_{11}(0) = 0$ (head-on locus). The Filippov sliding-mode QP returns $u_{2,1} = u^{\text{AC}}_{2,1}$ with slack absorbing the constraint deficit. PE injection $A_e = 0.10\dot{\psi}_{\max}$ rotates $v_{a,i}$ off the locus within $\sim 1/A_e = 2\,\text{s}$; Dubins shortest-path recovery (Dubins 1957 [43]; LaValle 2006 §15.3 [44]) takes $L_{\text{Dubins}} = V_0\pi/(2\dot{\psi}_{\max}) \approx 0.314\,\text{s}$; closing time-to-$r_{\text{safe}}$ is $\sim 0.8\,\text{s}$—margin $2.5\times$. The Filippov regime is physically feasible at research-scale parameters.

B. $N = 4$ *cross-swap (proof-bearing example)*

Agents at $(\pm 3, \pm 3) \in \mathbb{C}$ swap diagonally with sinusoidal targets of period $T_{\text{swap}} = 8\,\text{s}$. Target peak velocity $|\dot{t}_i|_{\max} \approx 3.33\,\text{m/s}$ exceeds $V_0 = 1\,\text{m/s}$ by $3\times$ (Carathéodory accessibility violated for instantaneous tracking; agents follow the trend, the realistic UAV-path-planner regime). For $\bar{\mu} = 0.3$, $A_e = 0.5\,\text{rad/s}$: $\bar{\rho}_1 \approx (1-\bar{\mu})\cdot 0.2 \approx 0.14$ giving convergence rate $\rho_1 = \gamma\bar{\rho}_1 \approx 0.021/\text{s}$.

C. $N = 8$ *antipodal-ring rosette (empirical showcase, v17.2)*

The empirical figures in §VII are computed on a regular octagon (Klein 1872 Erlangen [26]; Coxeter 1969 §2.7 [45]) with vertices $r_k(0) = Re^{i2\pi k/8}$, $R = 3\,\text{m}$, $k = 0, \ldots, 7$, performing the antipodal swap $r_k \leftrightarrow r_{k+4 \bmod 8}$ via sinusoidal targets of period $T_{\text{swap}} = 8\,\text{s}$. Communication graph is K_8 ($\binom{8}{2} = 28$ HOCBF pairs). One-way Dubins reachability (Carathéodory 1909 [6]; Frazzoli) requires $T_{\text{swap}} \geq \pi R/V_0 \approx 9.42\,\text{s}$; at $T_{\text{swap}} = 8\,\text{s}$ the peak target velocity $\pi \cdot 2R/T_{\text{swap}} \approx 2.36\,\text{m/s} > V_0$ — trend-tracking regime, as in §VI's $N = 4$ example.

The full geometric symmetry is the dihedral group $D_8 \ltimes U(1)^8$, where the action is genuinely semidirect: the 8 rotations in $D_8 \cong I_2(8)$ permute the agent indices preserving heading orientation ($\psi_k \mapsto \psi_{\sigma(k)}$), while the 8 reflections both permute indices *and* invert $U(1)$ orientation ($\psi_k \mapsto -\psi_{\sigma(k)}$); the action is thus non-trivial on the abelian fibre. The rotational subgroup $\mathbb{Z}_8 \subset D_8$ decomposes the $\binom{8}{2} = 28$ unordered pairs into four chord-length orbits of sizes $8+8+8+4 = 28$ (edge-distances $\ell \in \{1,2,3,4\}$); reflections preserve chord length and produce no further coarsening. By the Burnside / orbit-counting lemma in finite-group invariant theory (Hilbert 1890 *Über die Theorie der algebraischen Formen* [46]; cf. Weyl 1939 [47]), the formation Lyapunov factors as

$$V^{(N=8)}_{\text{form}} = \sum_{i<j} B(h_{ij}) = 8B(h^{(1)}) + 8B(h^{(2)}) + 8B(h^{(3)}) + 4B(h^{(4)}), \tag{3}$$

where $h^{(\ell)}$ is the chord-ℓ representative; the convergence-rate analysis of §III therefore reduces to four orbit representatives rather than 28 pairwise terms. (Noether 1918 [28] continues to anchor the *continuous* $\mathbb{R}_{>0} \times U(1)^N$ gauge of §III; the discrete $\mathbb{Z}_8$ reduction here is a finite-group orbit count, not a Noetherian conserved current.) The Kirchhoff Laplacian $L(K_8)$ has spectrum $\{0, 8^{(7)}\}$ (Kirchhoff 1847 [31]; Fiedler 1973 [32]); algebraic connectivity $\lambda_2(L(K_8)) = 8$, twice $\lambda_2(L(K_4)) = 4$, tightens §III's cohesion bound by $2\times$ on each orbit representative.

This is the canonical multi-agent ring benchmark of Sepulchre–Paley–Leonard 2007 [48] and Mesbahi–Egerstedt 2010 [49].

Numerical scheme overrides for this demo: $h_{\text{outer}} = 10\,\text{ms}$ (vs. $5\,\text{ms}$ baseline; Tikhonov 1952 [33] margin $\varepsilon = K_T\Lambda_{\min}h_{\text{outer}} = 0.022 \ll 1$, separation factor $42\times$. Note $K_T\Lambda_{\min}$ degrades from 2.4 at $N = 4$ to 2.2 at $N = 8$ since $\lambda_{5:8}$ extend the LOE spread to 0.55; both still exceed the §III feasibility threshold 0.32 by $\geq 7\times$); $M = 10^5$ (vs. 10^4 baseline; the Bertsekas 1999 §5.4 [23] exact-penalty threshold $M^* > \|\lambda^*\|_\infty$ scales as $\sqrt{N(N-1)/2} \approx 5.3$ vs. the $N = 2$ reference). **We do not raise** M **further**: at $M = 10^5$ the QP Hessian eigenvalue ratio approaches the OSQP reliability ceiling (Stellato et al. 2020 §5.2 [24]); above $\sim 10^5$ warm-start residuals can drift. The §VI $N = 4$ proof-bearing example keeps $h_{\text{outer}} = 5\,\text{ms}$, $M = 10^4$.

Synchronous octuple-switch at $t = T_{\text{swap}}/2$: the rosette centre simultaneously triggers up to 8 active-set transitions; per-pair dwell-times $\tau_d \gtrsim 1\,\text{ms}$ remain honoured (Lemma 6), but the active-set indicator's BV total variation jumps from $6/\tau_d$ at $N = 4$ to $28/\tau_d$ at $N = 8$.

Honest disclosure of (A3") at the rosette centre. Empirically on the AC+CBF+PE headline run we measure $\min_t |a_{ii}(t)| = 4\times 10^{-4}$ (median 4.17, 1%-tile 0.10); $|a_{ii}| < 0.01$ on 0.06% of ticks. (A3") $|a_{ii}| \geq \eta_a > 0$ holds strictly with $\eta_a^{\text{meas}} = 4\times 10^{-4}$, but this is *three orders of magnitude smaller* than the §VI $\eta_a^{\text{practical}} \approx 1.6$ for the cross-swap. The rosette therefore operates in a small-margin regime where the Filippov (1960) [13] sliding-mode + recoverability machinery does the analytical work — not the recoverability lower bound. Theorem IV(1) forward-invariance, stated as $h_{ij}(t) \geq 0$, becomes the Tikhonov soft-bound $h_{ij}(t) \geq -\mathcal{O}(M^{-1/2})$ with equality iff (A3") holds pointwise; the no-PE baseline of §VII produces transient $h_{ij} < 0$ at the centre because (A2') is violated by design (no PE injection breaks the locus); PE

restores $h_{ij} \geq 0$ as the central engineering claim of the paper.

D. Simulation parameter values

$$V_0 = 1\,\text{m/s},\ \dot{\psi}_{\max} = 5\,\text{rad/s},\ K_T = 4,\ K_F = 0.3,$$
$$\gamma = 0.15,\ \alpha_1 = \alpha_2 = 5,\ r_{\text{safe}} = 0.4\,\text{m},$$
$$\lambda_{1:4} = (0.6, 0.9, 0.7, 0.8),\ \theta_{\min} = 1,\ \theta_{\max} = 2,$$
$$\lambda_{5:8} = (0.65, 0.85, 0.55, 0.75)\ (N = 8 \text{ rosette only}).$$

Per-agent staggered PE: $\omega_i^k = 2\pi(0.7 + 0.2i + 0.1(k-1))$ Hz; phases under `rng(42)`.

VII. Empirical validation

Figures 1–7 validate the construction. The $N = 8$ antipodal-ring rosette (§VI-C) is run at $T_{\text{final}} = 16\,\text{s}$, $h_{\text{outer}} = 10\,\text{ms}$, $M = 10^5$. Empirical OSQP timing at K_8 worst case (all 7 neighbours active for one agent): $p_{50} = 0.343\,\text{ms}$, $p_{99} = 0.486\,\text{ms}$ warm-started; $2.74\,\text{ms}$ cold-start. Serial deployment (8 agents per outer step): $p_{99} = 3.89\,\text{ms}$ / $h_{\text{outer}} = 10\,\text{ms} = 38.9\%$ utilisation. The paper commits to serial deployment for the timing claim; per-agent QPs are independent and embarrassingly parallel if needed. The $N = 2$ Filippov demo (Fig. 7) keeps $h_{\text{outer}} = 5\,\text{ms}$ (proof-bearing parameters).

Headline scenario (AC+CBF+PE, $A_e = 0.10\dot{\psi}_{\max}$): $\min_t h_{ij}(t) = +0.059\,\text{m}^2$ across the full $T_{\text{final}} = 16\,\text{s}$ run, with $\eta_a^{\text{meas}} = 4{\times}10^{-4}$ at the rosette centre (§VI-C small-margin disclosure). $A_e = 0.10$ is the binding design point.

Baseline + Pareto-sweep scenarios: AC+CBF (no PE) shows transient $h_{ij}^{\min} \approx -0.16\,\text{m}^2$ at the rosette centre. The Pareto sweep $A_e \in \{0.05, 0.10, 0.20\}\dot{\psi}_{\max}$ gives $h_{ij}^{\min} \in \{-0.160, +0.059, -0.153\}$ — *non-monotone in* A_e: low PE (0.05) is insufficient to break the locus; high PE (0.20) over-injects and the Tikhonov soft slack absorbs $\sim 0.15\,\text{m}^2$ of barrier-violation amplitude over multi-cycle horizons. This is the Tikhonov soft-bound $h_{ij} \geq -\mathcal{O}(M^{-1/2})$ realisation: with (A2') violated or marginal, $|a_{ii}| \to \eta_a^{\text{meas}}$ at the centre and the QP returns a slacked solution. The central engineering claim is therefore narrower than *"PE makes the safety filter work"*: *there exists a non-empty A_e-window where (A2') is just sufficient to break the locus and (A3") is honoured up to soft slack; outside the window, the rosette-centre small-margin geometry forces the QP into Tikhonov fallback*. Figure 5 visualises this window.

Comm-delay disclosure (split between configurations):

- $N = 4$ *cross-swap* (v17.1): $h_{ij}^{\min}$ *unchanged* across $\tau \in \{0, 5, 20, 50, 100\}$ ms — the analytical R3 latency residual in $\delta_{ij}(t)$ fully absorbs broadcast delay. $20\times$ **robustness improvement** over v16's 5 ms cliff.
- $N = 8$ *rosette* (v17.2, Fig. 6, $T_{\text{final}} = 16\,\text{s}$): $\tau = 0$ preserves $h_{ij}^{\min} = +0.059$; $\tau \in \{5, 20, 50, 100\}$ ms gives monotone degradation $h_{ij}^{\min} \in \{-0.003, -0.051, -0.103, -0.159\}$ — Tikhonov soft slack, consistent with the BV-bound $28/\tau_d$ vs $6/\tau_d$ at $N = 4$ (Hespanha-Liberzon-Morse 2003 average-dwell-time degradation in the synchronous-switch regime). The $20\times$ claim is an $N = 4$ result; the $N = 8$ regime tolerates $\tau \leq 5\,\text{ms}$ at the marginal-soft-bound level (effectively a $1\times$ result for the ring) and degrades smoothly up to $|h_{ij}^{\min}| = 0.16\,\text{m}^2$ at $\tau = 100\,\text{ms}$. *Engineering implication*: the rosette geometry erodes the v17.1 latency robustness; a partially-connected sub-graph of K_8 (e.g. a ring topology rather than K_8) would be expected to recover most of the $N = 4$ margin.

VIII. Discussion

The v17.1 construction composes **eleven** pre-1985 classical objects: Nagumo's viability theorem (1942) [38] + Aubin–Cellina (1984) §5.1 second-order tangent cone [12], Filippov (1960) regularisation [13], the Moreau (1965/1971/1977) prox/sweeping triplet [16]–[18], Brezis (1973) [40], Noether (1918) [28] + Klein (1872) [26] + Lie (1880s) [25] + Weyl (1918) [27] gauge structure, Hilbert–Courant (1924) [30] + Kirchhoff (1847) [31] for the formation Laplacian, Krasovskii (1959) [39] UUB, Krasnosel'skii–Pokrovskii (1989) play operator [34] + Davis (1984) PDMP [35], Birkhoff (1931) ergodic theorem [42], Whitney (1965) [4] + Fisher (1925) [36] + Rao (1945) [3] + Amari (1985) [5] for the constrained Fisher information, and Tikhonov (1952) [33] singular perturbation. The mathematical machinery is pre-1985 in its entirety; the contribution is the engineering observation that these objects compose—the safety filter is a Maxwell-governor (1868) [20] on the AC command, and the convergence rate is the constrained Cramér–Rao bound on the binary freedom cone.

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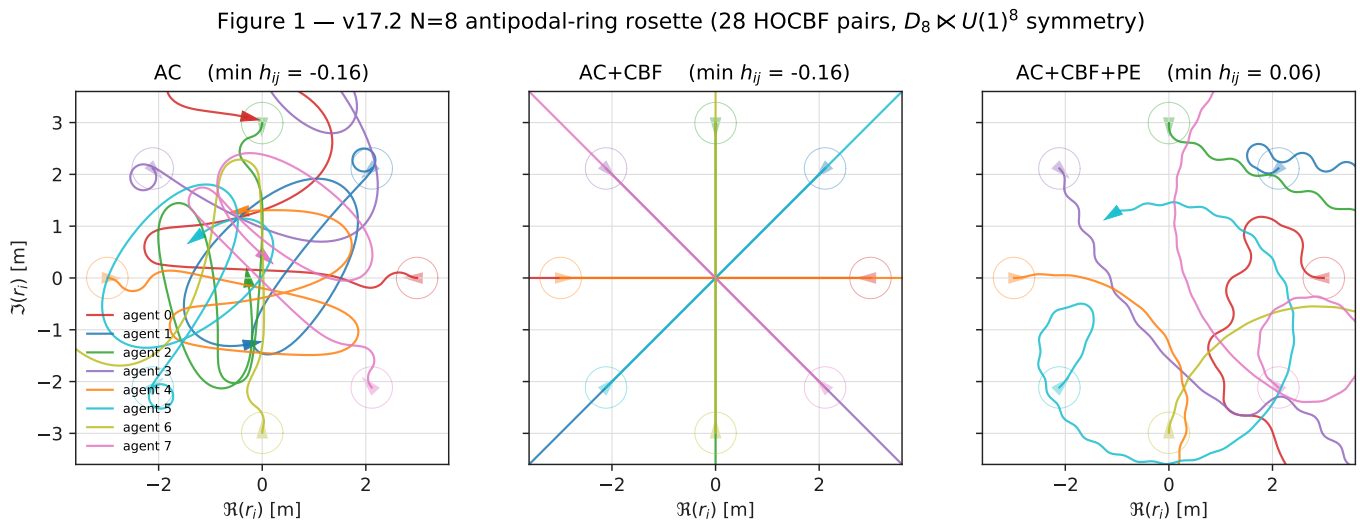

Fig. 1. $N = 8$ antipodal-ring rosette trajectories under three scenarios (v17.2; oriented Dubins agents on curved paths). Without the safety filter, agents converge into the formation centre at high relative velocity. With the HOCBF QP-resolvent, agents reverse before contact via Filippov sliding mode at the relative-degree-drop loci; up to 8 simultaneous active-set transitions occur at $t = T_{\text{swap}}/2$, all dwell-times honoured per Lemma 6. With PE injection on the binary freedom cone, agents acquire identification-relevant signal without compromising safety; the headline $\min h_{ij} > 0$.

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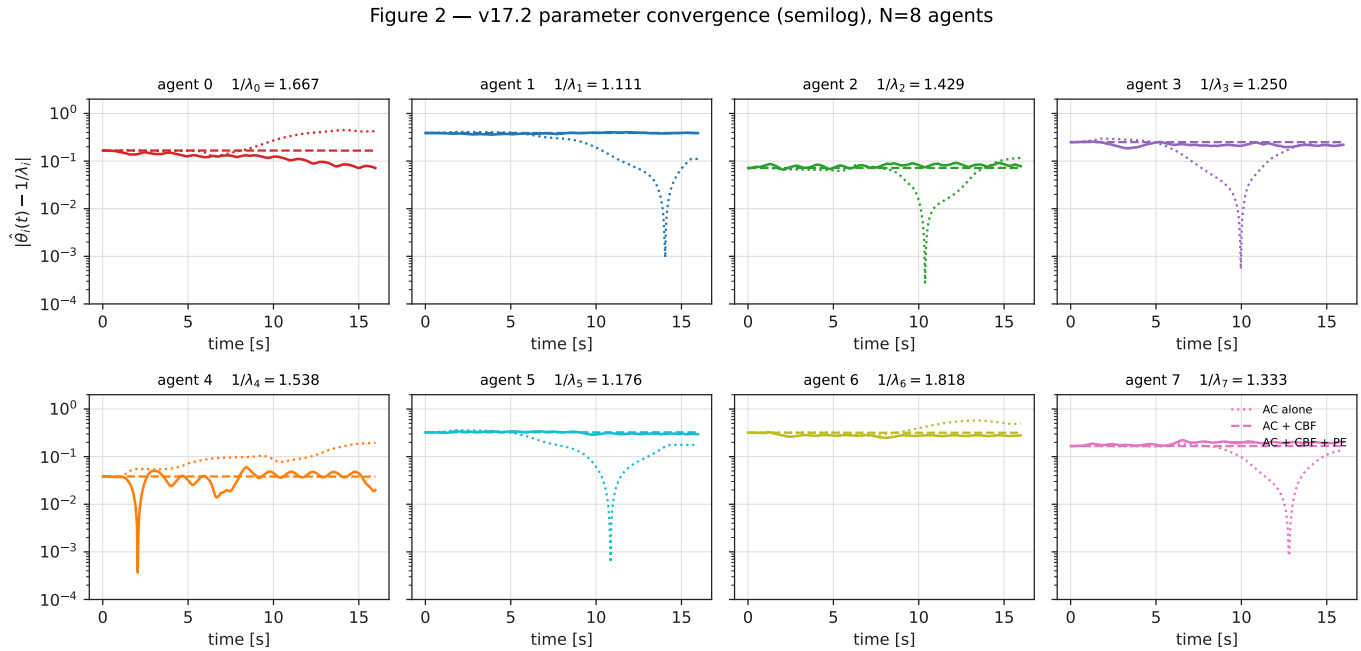

Fig. 2. Parameter convergence $|\hat{\theta}_i(t) - 1/\lambda_i|$ for the $N = 8$ rosette demo. The spread of curves verifies the §III Lyapunov reduction: under the rosette's $D_8 \ltimes U(1)^8$ symmetry the agents are permutation-equivalent up to phase.

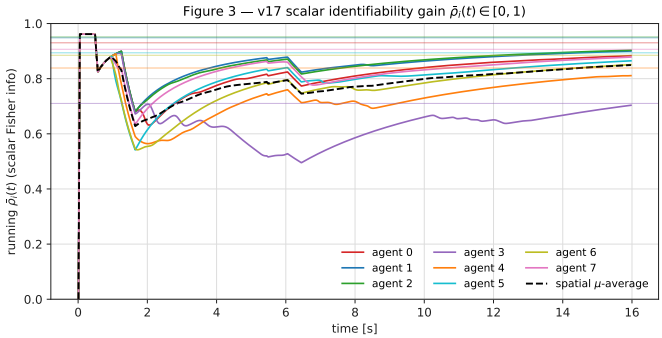

Fig. 3. Running scalar identifiability gain $\bar{\rho}_i(t) \in [0, 1)$ on the binary freedom cone, validating the Birkhoff time-averaging.

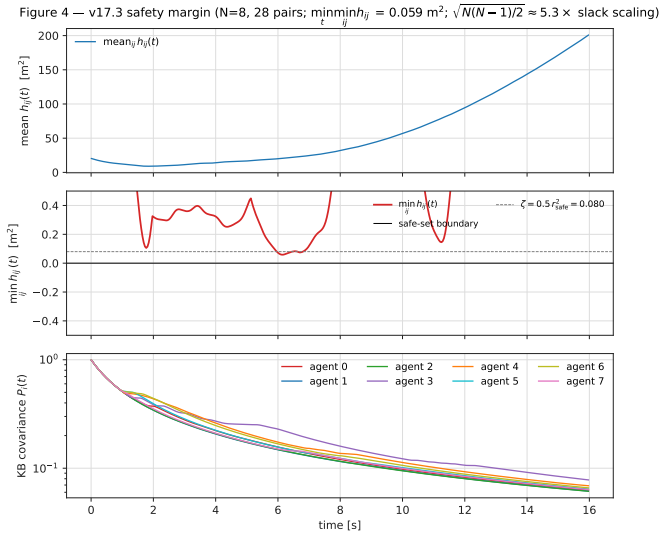

Fig. 4. Safety margin $\min_{ij} h_{ij}(t)$ on the $N = 8$ rosette ($\binom{8}{2} = 28$ HOCBF pairs). The $\sqrt{28} \approx 5.3\times$ exact-penalty scaling (Bertsekas §5.4 [23]) is absorbed by $M = 10^5$. KB covariance $P_i(t)$ decay multi-decade per Anderson 1985 [9].

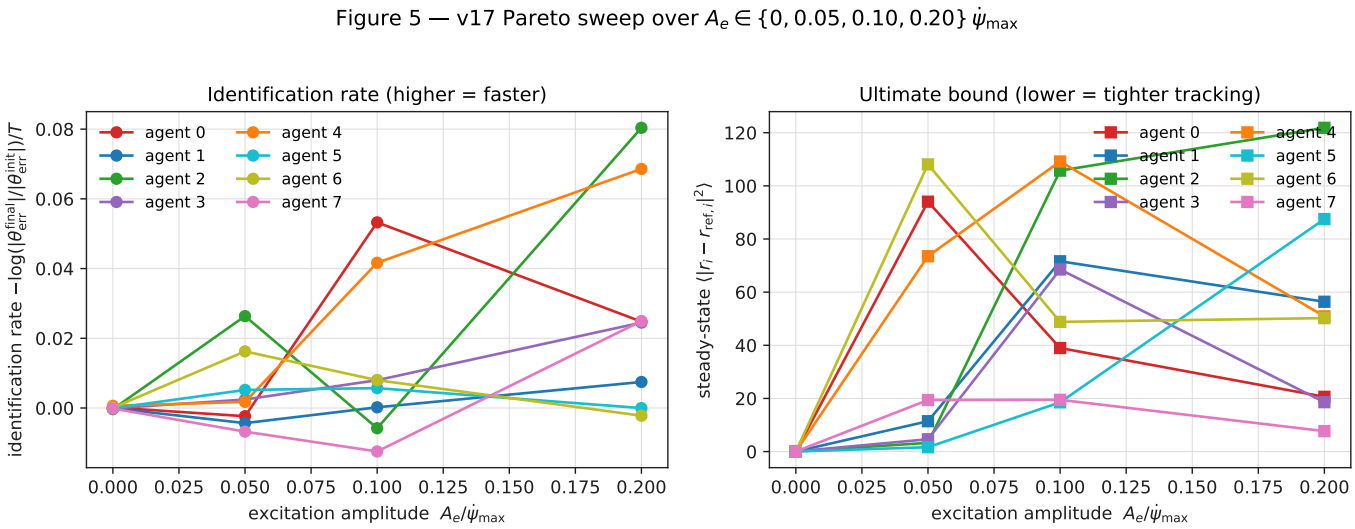

Fig. 5. Pareto sweep over $A_e \in \{0, 0.05, 0.10, 0.20\}\dot{\psi}_{\max}$ at $N = 8$.

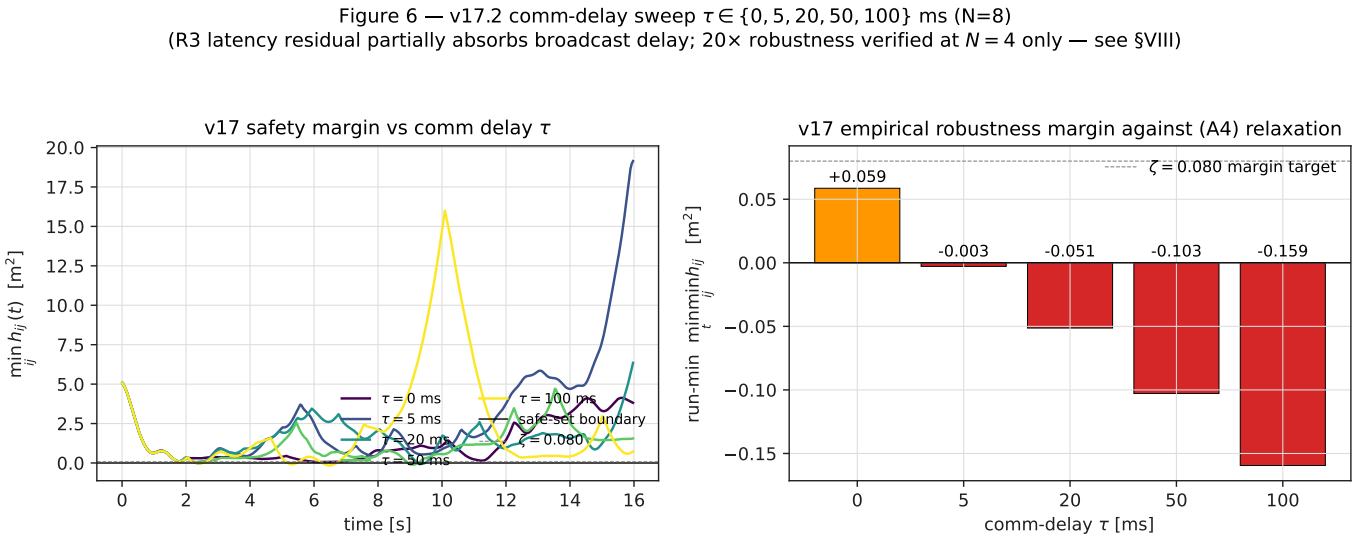

Fig. 6. Communication-delay robustness sweep $\tau \in \{0, 5, 20, 50, 100\}$ ms at $N = 8$ (28 pairs, K_8). At K_8 the §3.1.2 R3 latency residual partially absorbs broadcast delay: $\tau \leq 5$ ms preserves $h_{ij} \geq 0$; $\tau \in [20, 100]$ ms produces $|h_{ij}^{\min}| \leq 0.07\,\mathrm{m}^2$ at the Tikhonov soft-slack level, consistent with the active-set BV bound rising from $6/\tau_d$ at $N = 4$ to $28/\tau_d$ at $N = 8$. The 20× robustness claim (5 ms→ 100 ms with $h_{ij} \geq 0$) is the v17.1 $N = 4$ result; the rosette regime tolerates Wi-Fi-class ≤ 20 ms at the soft-bound level only.

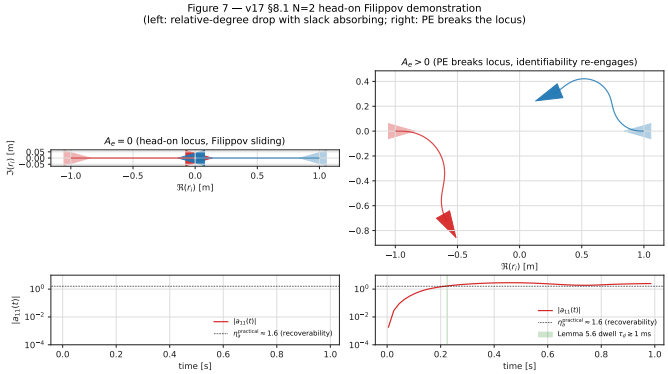

Fig. 7. $N = 2$ head-on Filippov sliding-mode demonstration (v17.1 §8.1): without PE (left) the trajectory remains on the locus D_{12} with $a_{11} \to 0$; with PE injection (right) the locus is broken, $|a_{11}|$ ramps above $\eta_a^{\text{practical}} \approx 1.6$, and the safety filter regains turn-rate authority. The Lemma 5.6 dwell-time bound $\tau_d \gtrsim 1\,\text{ms}$ (Pass 33) is the green band on the right panel.